

SYNTHEX

An Intelligent Literature-to-Materials Data Platform

Final Year Project Overview for Supervisor Review

Synthex turns research papers into structured, searchable and scientifically traceable materials data - while keeping the original evidence attached to every extracted claim.

Why Synthex matters

Materials researchers spend enormous time reading papers, copying values from tables, comparing processing conditions, and checking where each result came from. **Synthex automates this workflow without reducing a paper to untraceable numbers.**

What Synthex can now do

Capability	What it means in practice
Research-paper understanding	Reads scientific PDFs and routes them to the appropriate scientific domain instead of relying only on keyword matching.
Structured data extraction	Extracts materials, synthesis and fabrication steps, experimental conditions, performance metrics and computational descriptors into validated JSON records.
Native table intelligence	Reads tables as structured rows and columns, preserving headers, units, cell locations and table provenance rather than treating tables as mangled text.
Figure intelligence	Detects figures, captions, panels, axes, legends and annotations. It can also support controlled graph digitization while marking digitized values as estimated.
OCR fallback	Handles image-heavy or poorly encoded PDFs by using OCR only where native text extraction is insufficient, while recording that the evidence came from OCR.
Scientific provenance	Links extracted claims back to source pages, text snippets, tables, cells, figures or OCR regions so results remain auditable.
Ownership & review protection	Distinguishes focal experimental work from cited prior work, review summaries, examples and background information. Non-focal numerical claims are quarantined instead of silently entering the trusted archive.
Search-assisted enrichment	Can use web search for bibliographic discovery, DOI recovery and supplementary-information discovery without treating search snippets as scientific evidence.

Scientific domains

Gas Sensing

Extracts material preparation, device/fabrication details, gas-sensing conditions, response, sensitivity, selectivity, response/recovery time, LOD and relevant DFT descriptors.

Battery Materials

Handles synthesis, electrode fabrication, cell assembly, electrochemical testing, capacity, Coulombic efficiency, cycle life, retention, EIS-related values and battery DFT descriptors.

Catalysis & Electrocatalysis

The next scientific vertical is being built on the same validated core. The architecture is prepared for catalyst synthesis, reaction conditions, conversion/selectivity, activity, TOF, overpotential, Tafel slope, Faradaic efficiency, stability and computational catalysis descriptors.

Expandable Platform

The same architecture is designed to extend to corrosion, photovoltaics, thermoelectrics, membranes, additive manufacturing, semiconductors, biomaterials and other materials-science domains.

How the system works

Research PDF	Text + Tables + Figures	Domain Routing	Scientific Extraction	Evidence + Ownership Checks	Admission / Quarantine	Searchable Archive
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What makes Synthex different

- It does not merely summarize papers; it converts them into machine-readable scientific records.
- It preserves the distinction between reported values, normalized values, derived values, OCR-extracted values and digitized estimates.
- It separates experimental evidence from computational/DFT evidence instead of mixing both into one performance record.
- It keeps provenance and scientific ownership attached to data, making the resulting archive suitable for later comparison, visualization, knowledge-graph construction and model development.
- Its long-term aim is a literature-derived materials intelligence layer that complements established computational materials databases by capturing what researchers actually synthesize, test and report.

Current project status: Gas sensing and battery extraction are operational; Visual Intelligence, provenance, routing and admissibility are integrated into the shared core; Catalysis/Electrocatalysis is the next domain module being added.

In one sentence: Synthex is evolving from a paper reader into a traceable materials-science intelligence platform.